

Question 1 : Count Non-repeated Character in String

A data compression software utilizes various steps to compress a string of data. One of the steps involves finding the count of characters that are not repeated in the string.

Write an algorithm for the software developer to find the count of characters that are not repeated in the string.

Input Format

The input consists of a string.

compString representing the string to be compressed.

Output

Print an integer representing the count of characters that are not repeated in the string. If no such character is found or the input string is empty then print 0.

Note

The input string compString is case sensitive. Uppercase characters and lowercase characters are counted as different. The input string compString consists of alphanumeric and special characters only.

Example

Input: alphaadida

Output: 4

Question 2 : Difference Between the Count of Odd numbers and Even numbers

Write a program to return the difference between the count of odd numbers and even numbers.

Note : You are expected to write code in the **countOddEvenDifference** function only which will receive the first parameter as the number of items in the array and second parameter as the array itself. you are not required to take input from the console.

Example

Finding the difference between the count of odd and even numbers from a list of 8 number

Input1 : 8

Input2 : 10 20 30 40 55 66 77 83

Output: -2

Question 3 : Write a program to calculate and return the sum of absolute difference between the adjacent number in an array

Write a program to calculate and return the sum of absolute difference between the adjacent number in an array of positive integers from the position entered by the user.

Note : You are expected to write code in the **findTotalSum** function only which receive three positional arguments:

1st : number of elements in the array

2nd : array

3rd : position from where the sum is to be calculated

Example

Input:

Input 1 : 7
Input 2 : 11 22 12 24 13 26 14
Input 3 : 5

Output: 25

Question 4 : Write a program to find the difference between the elements at odd index and even index.

Write a program to calculate and return the sum of absolute difference between the adjacent number in an array of positive integers from the position entered by the user.

Note : You are expected to write code in the **findDifference** function only which receive the first parameter as the numbers of items in the array and second parameter as the array itself. You are not required to take the input from the console.

Example

Finding the maximum difference between adjacent items of a list of 5 number

Input:

Input 1 : 7
Input 2 : 10 20 30 40 50 60 70

Output: 40

Question 5 : Write a program for Decimal to Binary Conversion

Given a decimal number as input, we need to write a program to convert the given decimal number into an equivalent binary number.

Example

Convert 17 and 33 into it's respective decimal form

Input:

Input 1 : 17
Input 2 : 33

Output:

Output1 : 111
Output2 : 100001

Question 6 : Find a pair with maximum product in array of Integers

Given an array with both +ive and -ive integers, return a pair with the highest product.

Input:

Input1 : arr[] = {1, 4, 3, 6, 7, 0}
Input2 : arr[] = {-1, -3, -4, 2, 0, -5}

Output:

Output1 : {6,7}
Output2 : {-4,-5}

Question 8 : Find Total Feet

Given an array of integers representing measurements in inches, write a program to calculate the total of measurements in feet. Ignore the measurements that are less than a foot (eg. 10).

Note: You are expected to write code in the findTotalFeet function only which will receive the first parameter as the number of items in the array and the second parameter as the array itself. You are not required to take input from the console

Example:

Finding the total measurements in feet from a list of 5 numbers

Input:

5

18 11 27 12 14

Output:

5

Question 9: Largest Element in the Array

Problem statement 1

Given an array 'arr' of size 'n' find the largest element in the array.

Example:

Input: 'n' = 5, 'arr' = [1, 2, 3, 4, 5]

Output: 5

Explanation: From the array {1, 2, 3, 4, 5}, the largest element is 5.

Question 10: Problem statement

You have been given an array 'a' of 'n' unique non-negative integers.

Find the second largest and second smallest element from the array.

Return the two elements (second largest and second smallest) as another array of size 2.

Example :

Input: 'n' = 5, 'a' = [1, 2, 3, 4, 5]

Output: [4, 2]

The second largest element after 5 is 4, and the second smallest element after 1 is 2.

Question 11: Problem statement

You have been given an array 'a' of 'n' non-negative integers. You have to check whether the given array is sorted in the non-decreasing order or not.

Your task is to return 1 if the given array is sorted. Else, return 0.

Example :

Input: 'n' = 5, 'a' = [1, 2, 3, 4, 5]

Output: 1

The given array is sorted in non-decreasing order; hence the answer will be 1.

Question 12. Problem statement

You are given a sorted integer array 'arr' of size 'n'.

You need to remove the duplicates from the array such that each element appears only once.

Return the length of this new array.

Note:

Do not allocate extra space for another array. You need to do this by modifying the given input array in place with $O(1)$ extra memory.

For example:

'n' = 5, 'arr' = [1 2 2 2 3].

The new array will be [1 2 3].

So our answer is 3.

Question 13. Given an array 'arr' containing 'n' elements, rotate this array left once and return it.

Rotating the array left by one means shifting all elements by one place to the left and moving the first element to the last position in the array.

Example:

Input: 'a' = 5, 'arr' = [1, 2, 3, 4, 5]

Output: [2, 3, 4, 5, 1]

Explanation: We moved the 2nd element to the 1st position, and 3rd element to the 2nd position, and 4th element to the 3rd position, and the 5th element to the 4th position, and move the 1st element to the 5th position.

Question 14. Problem statement

Given an array 'arr' of 'n' non-negative integers, your task is to move all the zeros to the end of the array while keeping the non-zero elements at the start of the array in their original order. Return the modified array.

Example :

Input: 'n' = 5, 'arr' = [1, 2, 0, 0, 2, 3]

Output: [1, 2, 2, 3, 0, 0]

Explanation: Moved all the 0's to the end of an array, and the rest of the elements retain the order at the start.

Question 15. Maximum consecutive ones in array

Problem statement

You are given a sorted array 'arr' of positive integers of size 'n'.

It contains each number exactly twice except for one number, which occurs exactly once.

Find the number that occurs exactly once.

Example :

Input: 'arr' = {1, 1, 2, 3, 3, 4, 4}.

Output: 2

Explanation: 1, 3, and 4 occur exactly twice. 2 occurs exactly once. Hence the answer is 2.
